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Applied Machine Learning
Class Project 01: Classification
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Human Activity Recognition Using Smartphone Sensor Data

1. Dataset Description

For this project, I will use the Human Activity Recognition Using Smartphones dataset from the UCI Machine Learning Repository. The dataset contains 10,299 observations collected from 30 volunteers aged 19–48 who performed six everyday physical activities while wearing a Samsung Galaxy S II smartphone around their waist.

The dataset includes 561 numerical features extracted from accelerometer and gyroscope measurements. These features represent movement patterns, acceleration, angular velocity, and statistical characteristics of the recorded signals. The data is divided into 7,352 training observations and 2,947 testing observations, exceeding the project's minimum requirement of 10,000 total data points.

Dataset source: https://archive.ics.uci.edu/dataset/240/human+activity+recognition+using+smartphones

2. Project Objectives

The primary objective of this project is to develop and evaluate machine learning classification models that can recognize human physical activities using smartphone sensor data.

Specifically, I aim to classify six activities:

- Walking
- Walking upstairs
- Walking downstairs
- Sitting
- Standing
- Lying down

I will implement and compare four classification algorithms: K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest, and Gaussian Naive Bayes.

I will evaluate their performance using accuracy, precision, recall, macro F1-score, and confusion matrices. My goal is to identify which algorithm provides the best balance of classification performance and computational efficiency.

3. Significance of the Project

Human Activity Recognition is an important application of machine learning that connects computer science with human-computer interaction and healthcare technology.

Smartphones and wearable devices collect movement data that can be used to recognize physical activities. These capabilities have potential applications in fitness monitoring, rehabilitation technology, assistive systems, and activity-aware interfaces.

I am particularly interested in understanding how machine learning can interpret human behavior through sensor data and how similar techniques might support healthcare-related technologies.

This project will allow me to explore practical applications of classification algorithms while developing skills in sensor-data processing, model evaluation, and human-centered machine learning.

4. Potential Limitations and Challenges

Although the dataset contains more than 10,000 observations, it was collected from only 30 participants. This limits how confidently the results can be generalized to a larger and more diverse population.

Another challenge is the high dimensionality of the dataset, with 561 features that may contain redundant or highly correlated information. Some activities, particularly sitting and standing, may also have similar movement patterns, making classification more difficult.

I will address these challenges by examining feature distributions, applying appropriate preprocessing and scaling, and comparing multiple classification algorithms.

I will preserve the dataset's original participant-separated training and testing sets to reduce the risk of data leakage. Hyperparameter tuning will use training data only.

The dataset also represents controlled activities performed with a smartphone positioned at the waist, so the results may not directly apply to other devices or real-world conditions.

5. Project Timeline

Week 1: October 23–29

- Review the dataset documentation and explore its features and activity labels.
- Examine data quality, class distributions, and preprocessing requirements.
- Implement initial versions of the four classification algorithms.

Week 2: October 30–November 5

- Evaluate initial models and prepare the technical progress report.
- Tune selected hyperparameters and compare model performance.
- Analyze confusion matrices and classification errors.
- Select the best-performing model and develop a simple interactive demonstration.
- Prepare the final presentation and document the results.

Expected Outcome

By the end of this project, I expect to have a working machine learning classification system capable of predicting six physical activities from smartphone sensor features.

I will present a comparison of four classification algorithms, explain the strengths and limitations of each approach, and demonstrate the selected model using previously unseen test observations.

The project will provide practical experience with a complete machine learning workflow, from dataset exploration and preprocessing to model training, evaluation, and implementation.